

# Aaryan Naruka

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## EDUCATION

- **Michigan State University Honors College** – East Lansing, MI
- Bachelor of Science in Computational Data Science | GPA: **3.8** | Class of 2026

## PROJECTS

### Autonomous Car | September 2023 – May 2024

- Created a traffic light detection program by creating bounding boxes around the lights with YOLOv8 ● Examined and found the biggest bounding box as well as best (most straight forward) angle to determine the best light to choose for the sensors to focus on ([YouTube Demo](#))
- Utilized RoboFlow to annotate and preprocess images of parking lots, training a YOLOv8 model to accurately distinguish white parking spots from image data with high precision

### Sports Analyst – Premier League Predictor | September 2024 – December 2025

- Data cleaning which made data viable for regression and modeling in programs like TensorFlow and PyTorch exploring NLP techniques and Big Data pipelines for automated data extraction and analysis. ● Conducted trend analysis on team performance (wins, losses, goals, and defensive metrics for teams in the last 21 years) to identify Manchester City as the top contender, with 80%-win consistency since 2017
- Designed advanced visualizations (stacked bar charts, dynamic line graphs) to communicate insights effectively to technical and non-technical viewers (**pandas, NumPy, matplotlib, seaborn**)

### Forest Fire Model – Agent-Based Modeling

- Built a model to simulate forest fire dynamics, using NumPy to represent a 2D forest grid and implement state transitions for empty, tree, and on fire cells
- Utilized Matplotlib to create real-time visualizations and animations of fire propagation, integrating efficient rendering of dynamic grid states for large-scale simulations
- Extended model functionality to support multiple fire initiation points and boundary conditions, analyzing their impact on fire spread dynamics and tipping point shifts ([GitHub Repo](#))

## WORK EXPERIENCE

### Jackson National Life – Lansing, MI

#### Data Science Intern & Strategic Support Associate | May 2025 – Aug 2025

- Supported the **Enterprise Data Platform** in an **Azure cloud environment**, assisting with **automating and migrating** on-premises datasets to secure, scalable cloud-based infrastructure
- Curated **reports and dashboards in Power BI using Power Query/M and Dax**, connecting them to enterprise data sources to deliver actionable insights for stakeholders and presenting findings to both technical and non-technical audiences
  - Learned how to present data and analysis in a way that is easy to comprehend and follow up on
  - Built connections with professionals across departments including Finance, Accounting, and Sales ●

Created **SQL queries** to access JNL databases and prepare data sources for analytical use

- Gained hands-on training in **cloud data platforms, BI tools, and enterprise analytics**, expanding technical skills in a professional financial services environment
- Applied **cybersecurity principles** to practice safe data extraction and usage
  - Followed **data governance policies** to ensure compliance with internal standards and industry regulations in the financial services sector

### Reliable Therapy Services – Troy, MI

#### Office Admin | May 2023 – Aug 2023, May 2024 – Aug 2024

- Automated the patient intake process by developing a program that streamlined data entry into Excel, ensuring accurate and accessible records for sharing across companies

## INVOLVEMENT

**MSU Artificial Intelligence Club** – Member (Sept 2022 – Present) **Society of Data Scientists** – Leadership Role (Sept 2023 – Present)

**MSU Tutors** – CSE, PHY, MTH Tutor (Sept 2023 – Present)

## TECHNICAL SKILLS

**Languages:** Python, R, C++, Java, SQL, HTML/CSS **Libraries/Frameworks:** TensorFlow, PyTorch, YOLOv8, Pandas, NumPy, Matplotlib, Seaborn, React, Scikit-learn, Big Data Pipelines (ETL), NLP (Text Analytics) **Tools:** Git, GitHub, VS Code, PyCharm, Xcode, Jupyter, Excel, Anaconda, Docker, Power BI (Power Query/M, DAX) **Platforms:** Linux, macOS, Windows, Cloud Computing, Azure Cloud

## COURSE WORK

Calculus I–IV | Probability & Statistics | Machine Learning (CSE 404) | Database Systems (CSE 480) | Data Structures & Algorithms (CSE 331) | C++ Programming (CSE 232) | Physics I–II | EGR 100 | Data Mining | AI & Deep Learning | Business Analytics & Forecasting